A machine-learning-based approach to expedite predicting effective metamaterial properties for electromagnetic transmission control

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ABSTRACT

Metamaterials are a new class of engineered materials capable of controlling electromagnetic (EM) waves in unique ways. They have applications in EM shielding, cloaking, and wave absorption. However, designing metamaterials with specific EM properties is often a time-consuming and computationally intensive process, relying heavily on iterative simulations and trial-and-error adjustments.

This study employs a machine learning (ML) framework to accelerate the prediction and optimization of metamaterial properties. Deep learning models are used to establish a relationship between geometric and material parameters with EM characteristics such as permeability, permittivity, and scattering parameters. A comprehensive dataset was generated using CST Studio Suite, containing over 1,100 metamaterial designs and their corresponding EM responses across the 20–32 GHz frequency range.

To streamline dataset generation, a custom Python API was developed to interface directly with CST, automating the simulation workflow and data collection. Furthermore, a backward prediction model based on a custom artificial neural network (ANN) architecture was introduced to generate new metamaterial designs from desired frequency responses. The proposed model achieved a training loss of 0.8433 and was validated through additional CST simulations, confirming its predictive accuracy.

Simulation and experimental results demonstrate that the proposed approach significantly reduces design time while maintaining high performance. Overall, the integration of ML into metamaterial design presents a promising path for rapid development of EM-controlling structures for applications in wearable electronics, radar systems, and next-generation 5G technologies.

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Chapter 1

Introduction

1.1 Research Background

Electromagnetic (EM) radiation management has become a critical concern in modern technology due to its wide-ranging impact on communication systems, electronic devices, and human health. Traditional materials used for EM shielding, such as conductive metals and ferrites, exhibit limitations in terms of weight, flexibility, and frequency selectivity. These challenges have led to the search for more efficient design methods.

Metamaterials are engineered structures that control electromagnetic (EM) waves through geometry rather than material composition. They can manipulate wave propagation at subwavelength scales. This makes them useful in antennas, cloaking, stealth, and EM shielding. Research in the last two decades has shown significant progress. Still, designing and optimizing metamaterials for specific goals, such as EM radiation reduction, remains difficult.

Traditional design relies on numerical simulations like the finite element method (FEM) or finite-difference time-domain (FDTD) method. These methods are accurate but require high computational resources and long processing times. Exploring large design spaces is slow and often limited. This restricts the discovery of optimal designs that balance efficiency, manufacturability, and stability.

Machine learning (ML) offers a faster alternative. ML models can learn the mapping between structure and EM response. Once trained, they predict results without running full simulations. Approaches such as deep neural networks, ensemble methods, and data-driven optimization have shown good performance for wave control, absorption, and shielding. However, many methods struggle with generalization, limited data efficiency, and weak physical interpretability.

This work develops an ML-based framework to predict effective metamaterial properties for EM transmission control. The system uses automated dataset generation and both forward and inverse models. It reduces design time and supports practical development of EM shielding structures.

1.2 Problem Statement

Metamaterials can effectively control electromagnetic (EM) radiation by shaping wave propagation. Their properties depend on many design parameters. Predicting these properties with traditional simulations is slow and computationally expensive. The geometry-to-response relationship is nonlinear, making it hard to identify optimal designs quickly.

These limitations hinder rapid development of optimized metamaterials. ML can help, but current models often require large datasets, generalize poorly across frequencies, or ignore key physical behaviors. In addition, there are no publicly available datasets that comprehensively cover a wide frequency range, such as 20–32 GHz, since most existing studies focus only on a single frequency band. This lack of diverse data limits model generalization and the ability to design broadband metamaterials. Moreover, generating backward designs—creating a metamaterial structure that performs optimally for a specific required frequency—remains a significant challenge due to the complex, non-invertible mapping between geometry and EM response.

A faster, more accurate ML framework is therefore needed. Such a framework would enable efficient prediction of metamaterial properties, support inverse design capabilities, and shorten design cycles, accelerating innovation in EM shielding materials and related applications.

1.3 Motivation

Electromagnetic radiation causes signal loss, poor device performance, and possible health effects. Metamaterials can reduce these issues by controlling how EM waves move and by improving shielding and absorption. Designing these materials, however, needs repeated simulations and testing. This process is slow and consumes many resources.

Machine learning has shown strong ability in speeding up material research. It learns patterns directly from data and reduces the need for large numbers of simulations. Applying ML to metamaterial design can make property

prediction faster and design optimization more practical. This work is driven by three main goals: to lower computational cost, to speed up metamaterial discovery, and to support applications in wireless systems, stealth, and electromagnetic compatibility (EMC).

1.4 Objectives

Develop Machine Learning Models: Train and test ML algorithms such as gradient boosting and neural networks for both forward (parameters → frequency) and inverse (frequency → parameters) prediction of metamaterial behavior.

Create a Structured Dataset: Generate a large dataset automatically using CST Microwave Studio. Record parameter changes and their S-parameter responses within the 20–32 GHz frequency range.

Optimize Metamaterial Designs: Use trained models to select designs that reduce EM radiation effectively.

Validate Predictions: Compare ML-based results with CST simulations to measure accuracy, consistency, and computation time.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The integration of machine learning (ML) into metamaterial research has emerged as a transformative approach, enabling faster, data-driven design processes that were previously constrained by the limitations of traditional analytical and simulation techniques. By leveraging ML models, researchers are now capable of accelerating the design cycle, improving predictive accuracy, and exploring vast parameter spaces with unprecedented efficiency. These advancements hold particular promise for applications in electromagnetic (EM) radiation reduction, wireless communication, stealth technology, biomedical devices, and next-generation sensing systems. Despite these

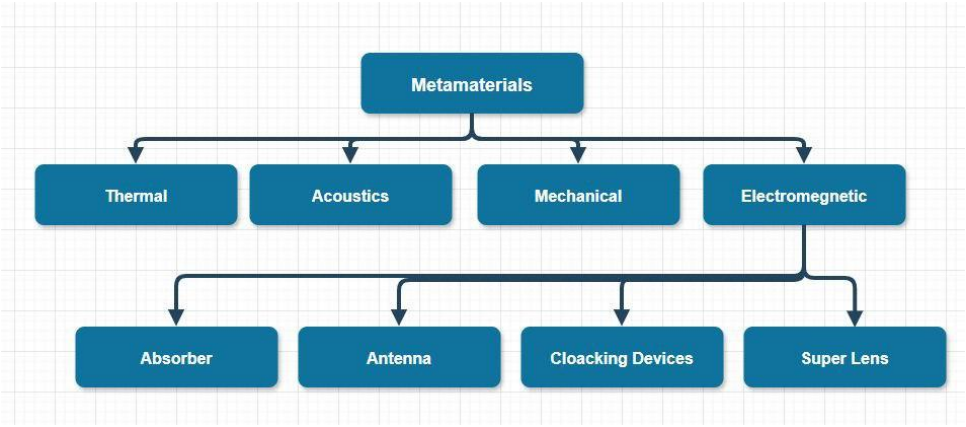


Figure 2.1: Types of metamaterials and their applications in electromagnetic domains

notable achievements, several critical challenges remain. Issues such as model scalability, interpretability of black-box predictions, and data efficiency limit the full adoption of ML in real-world metamaterial design. The integration of physics-informed ML models, which combine computational intelligence with domain knowledge, is emerging as a promising solution to bridge these gaps. This hybrid paradigm has the potential to not only enhance prediction accuracy but also provide design insights that

are physically interpretable and practically implementable.
The growing interest in ML-driven metamaterial design stems from the increasing demand for computationally efficient alternatives to conventional methods. Traditional numerical simulations such as the finite element method (FEM) and finite-difference frequency-domain (FDFD) remain accurate but are prohibitively time-consuming when navigating high-dimensional design spaces. In contrast, ML-based surrogate models offer rapid prediction and optimization capabilities, accelerating the development of high-performance metamaterials tailored to specific applications.

2.2 Scope of Research

The present research on 'Machine-Learning-Based Approach to Expedite Predicting Effective Metamaterial Properties for Electromagnetic Transmission Control" focuses on leveraging AI-driven frameworks to enhance both the efficiency and accuracy of metamaterial property prediction. The central motivation lies in addressing the bottlenecks associated with manual design and computationally heavy simulations, thereby enabling scalable solutions for industrial and academic use cases.
This study emphasizes the development of models that reduce computational costs, improve predictive reliability, and support inverse design tasks. By targeting these objectives, the research aims to provide contributions of both theoretical significance and practical utility. Furthermore, the scope extends to addressing broader challenges, such as improving generalization across frequency bands and material types, as well as identifying future research directions that ensure long-term relevance in an evolving technological landscape.

2.3 Review of Existing Literature

2.3.1 Traditional Approaches

The field of metamaterials originated with Veselago (1968), who introduced the concept of negative index materials, sparking decades of research into electromagnetic cloaking, perfect lenses, and absorbers. Early studies relied heavily on analytical models and numerical simulations to investigate wave–material interactions. While such methods provided high-fidelity insights, they became increasingly inefficient as design complexity and dimensionality grew. High computational costs limited their feasibility for large-scale design exploration and rapid prototyping. To mitigate these

issues, researchers introduced heuristic optimization techniques such as genetic algorithms (GA), particle swarm optimization (PSO), and topology optimization.
Moreover, designing effective metamaterials typically requires a repetitive trial-and-error process, where researchers manually adjust structural parameters, simulate each variation, and analyze the resulting electromagnetic response. This iterative workflow demands substantial time and computing power, especially when exploring large parameter spaces. Additionally, metamaterials often need to be tailored for specific frequency bands, meaning a design optimized for one frequency may not perform well at another. Consequently, new structures must be generated and simulated separately for each target frequency range, further increasing computational expense and design effort. These challenges collectively highlight the need for automated, data-driven design methodologies that can accelerate discovery and reduce dependence on conventional simulation-based optimization.

2.3.2 Machine Learning Approaches

The adoption of ML has significantly expanded the methodological landscape of metamaterial research. Early implementations demonstrated the potential of supervised learning in predicting metamaterial responses with remarkable efficiency. For example, Peurifoy et al. [17] applied deep neural networks (DNNs) to predict the optical properties of multilayer nanostructures, achieving orders-of-magnitude acceleration compared to conventional simulations. Similarly, Liu et al. [12] employed convolutional neural networks (CNNs) on structured datasets, demonstrating enhanced feature extraction for predicting EM responses.
Beyond deep learning, ensemble methods such as decision trees, random forests (RF), and Gaussian process regression (GPR) have been utilized to create surrogate models. Among these, GPR is particularly noteworthy for its ability to quantify prediction uncertainty, offering valuable guidance in experimental validation and adaptive design strategies.

2.3.3 Inverse Design Paradigm

One of the most challenging and impactful research areas is the inverse design problem: determining the optimal metamaterial structure that produces a desired EM response. Generative models have recently revolutionized this domain. Variational autoencoders (VAEs) and generative adversarial networks (GANs) have been employed to produce innovative

designs that were previously unattainable through manual methods. For instance, Zhang et al. [28] proposed a GAN-based framework that efficiently generated complex metamaterial configurations with superior absorption characteristics, showcasing the potential of data-driven generative techniques.

In addition, these generative approaches not only accelerate the design process but also enable the discovery of unconventional geometries that challenge traditional design intuition. Recent studies have shown that hybrid architectures, combining physics-based constraints with deep generative models, can further improve prediction accuracy and design feasibility. The integration of optimization algorithms such as genetic algorithms or reinforcement learning with GAN-based systems has also demonstrated promising results in refining generated patterns. Consequently, inverse design through generative modeling is emerging as a cornerstone technique for next-generation metamaterial innovation.

2.3.4 Recent Contributions in AI-Metamaterial Research

A summary of notable recent contributions is presented in Table 2.1. These studies highlight the diversity of ML techniques applied in the field, ranging from elite-preserving genetic algorithms for antenna design to interpretable random-forest-based inverse design models. Collectively, they demonstrate the rapid expansion of AI-driven metamaterial research and underscore the importance of scalable, interpretable, and application-driven frameworks. Furthermore, the reviewed works reveal a growing emphasis on hybrid methodologies that combine data-driven learning with physics-informed constraints, enabling more reliable generalization across frequency ranges and structural variations. Deep reinforcement learning and surrogate modeling approaches have also shown potential in adaptive design optimization, reducing dependency on extensive simulation datasets. As the field progresses, there is a clear trend toward integrating explainable AI techniques, ensuring that model predictions can be interpreted and validated by domain experts. These advancements collectively pave the way for a new generation of intelligent, autonomous design systems capable of revolutionizing metamaterial engineering. However, despite these achievements, the absence of standardized datasets and unified evaluation metrics continues to limit direct performance comparison and cross-model validation across studies.

Table 2.1: Selected studies on AI-driven metamaterial design: Focus, methods, and contributions.

Research Focus	**Methods**	**Key Contributions**
Sun et al. [26]: Automated optimization of near-zero refractive index metamaterials (NZIM) for high-directivity antennas	Elite-preserving genetic algorithm (EGA)	Demonstrated the potential of NZIM in achieving highly directional antenna designs
Mohammed et al. [14]: Absorption properties of metamaterial absorbers (MMA) across 20–26 GHz for 5G	Metamaterial absorber (MMA) design	Developed a five-band MMA achieving nearly perfect absorption with compact structure for 5G communication
Cerniauskas et al. [1]: Data-driven design for enhanced metamaterial performance	ANN, GA, PCA	Introduced dimensionality reduction for efficient exploration of design spaces in metamaterial optimization
Qian et al. [19]: Intelligent metasurfaces for advanced EM applications	Tandem neural networks, MLP, CNNs	Highlighted the role of intelligent metasurfaces in transforming computational methods and laboratory practices
Si et al. [23]: Integration of AI with artificial metamaterials for wave control	Supervised/unsupervised learning, CNNs	Showcased AI-driven forward prediction and inverse design in plasmonic metamaterial applications
Chen et al. [3]: Interpretable generative inverse design framework	Random-forest-based RIGID	Proposed RIGID for fast, interpretable metamaterial design without extensive training datasets
Lee et al. [7]: Generative design for structural metamaterials	FEA, ML-based generative design	Validated ML-enhanced designs with over 90% improvement in buckling strength while maintaining modulus

2.4 Summary and Research Gaps

The reviewed literature demonstrates the significant progress achieved in the design and optimization of metamaterials through both traditional and modern computational techniques. Traditional approaches, such as analytical modeling, FEM, and FDFD simulations, have provided deep physical insights but remain limited by computational inefficiency, particularly for large-scale or high-dimensional design problems. Optimization algorithms such as GA, PSO, and topology-based methods attempted to overcome these challenges but still required extensive iterative simulations, limiting their real-time applicability.

With the advent of machine learning, metamaterial research has entered a new phase. Supervised learning techniques (e.g., DNNs, CNNs, RF, and GPR) have proven effective in accelerating forward prediction of EM responses, while generative models (e.g., VAEs and GANs) have shown strong potential in solving inverse design problems. Recent studies further highlight the integration of AI with physics-informed models, interpretable frameworks, and hybrid generative approaches, enabling the development of designs with improved accuracy, adaptability, and reduced computational demands.

Despite these advances, several research gaps remain:

Scalability: Many ML models are restricted to narrow frequency ranges or specific design configurations, limiting their generalization across broader metamaterial classes.

Data Efficiency: The reliance on large, manually generated simulation datasets remains a major bottleneck. Developing methods that require fewer samples while maintaining accuracy is a pressing challenge.

Interpretability: Existing deep learning models often operate as black boxes, offering little physical intuition. More interpretable and physics-informed ML models are required for practical adoption in industry and academia.

Inverse Design Robustness: Although generative models show promise, they frequently suffer from issues such as mode collapse, instability, or lack of functional guarantees. Ensuring reliability in inverse design remains an open problem.

Experimental Validation: Much of the reported progress is based on simulations, with limited experimental verification. Bridging this gap is crucial to translate AI-driven designs into real-world applications.

In summary, the existing body of work establishes a strong foundation for ML-driven metamaterial research but leaves considerable room for improvement in terms of scalability, interpretability, and data efficiency. Addressing these challenges motivates the present study, which seeks to develop a machine-learning-based framework for predicting effective metamaterial properties aimed at electromagnetic radiation reduction. This approach is expected to not only reduce computational costs but also improve design reliability and contribute to the broader adoption of AI-assisted metamaterials in advanced technological applications.

Chapter 3

Methodology

3.1 Research Design

Fifth-generation (5G) communication systems rely on the millimeter-wave spectrum. Among these, 28 GHz is one of the most important carrier frequencies. This study focused on developing metamaterial unit cells that can resonate at 28 GHz. The design was also tested over a wider range of 20–32 GHz to ensure stable operation in practical conditions.

The design process started with the selection of main structural parameters. These included substrate size, copper thickness, resonator shape, and split-gap width. Initial models were created and tested using CST Microwave Studio. The S-parameter simulations helped to study resonance and transmission behavior. Based on the results, each design was modified and tested several times to reduce reflection and improve transmission.

After several iterations, an optimized unit-cell design was obtained. To study the effect of geometric variations, different parameter values were changed step by step. Each variation was simulated and analyzed to observe how these changes affected the frequency response. This process helped in understanding the connection between structure and electromagnetic behavior.

Finally, the designs were verified to ensure accurate performance. Resonance trends, effects of geometric changes, and consistency in repeated simulations were checked carefully. This structured method ensured that the final results followed physical principles and could be used reliably for practical metamaterial applications.

3.2 Tools and Technologies

CST Microwave Studio — primary tool for electromagnetic design and simulation.

Python — programming interface for automation and data processing.

pywin32 — library for establishing Windows COM communication with CST.

Custom Python API — developed to automate CST workflows (design creation, simulation execution, results extraction).

Python packages — including `numpy`, `matplotlib`, and `sqlite3` for numerical processing, visualization, and structured dataset storage.

Machine Learning Frameworks — tested various ML/DL methods (e.g., regression models, neural networks) for both forward and backward metamaterial prediction tasks.

3.3 Workflow

The overall workflow (Figure 3.1) integrates CST simulation with automated data generation and ML model training:

1. Define parameter values and design structure in Python.

2. Launch CST Microwave Studio via Python API.

3. Generate the unit-cell design and configure boundary conditions.

4. Run simulations over the frequency range (20–32 GHz).

5. Extract S-parameter results (focus: **SMaxMin**).

6. Store results and parameters in an SQLite database.

7. Preprocess the data into suitable format for ML model training.

8. Train ML models for forward and backward predictions.

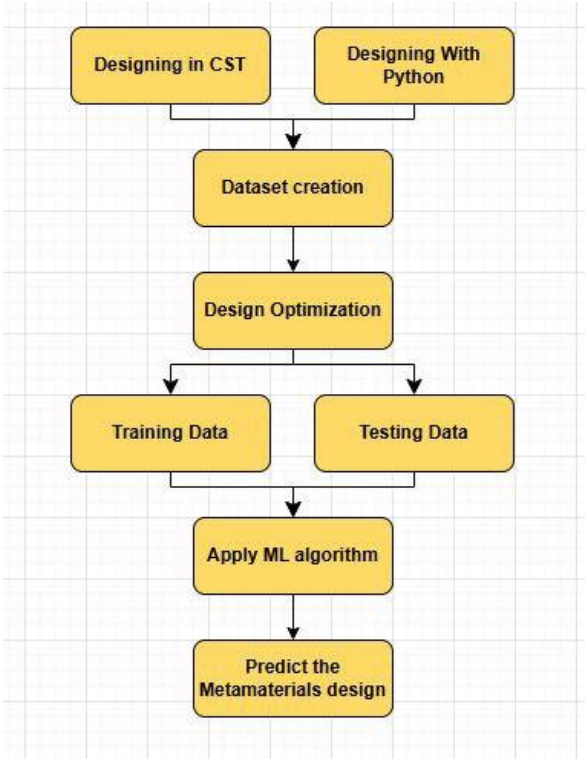


Figure 3.1: Overall methodology workflow integrating CST simulation with automated dataset generation and ML training.

3.4 System Design

3.4.1 Dataset Creation

To visually demonstrate the link between structure and response, three comparative examples are provided (Figures 3.2–3.4). Each figure contains:
Left panel: CST-generated unit-cell geometry.
Right panel: Corresponding S-parameter response (plot of $|S_{11}|$ and $|S_{21}|$ vs frequency).

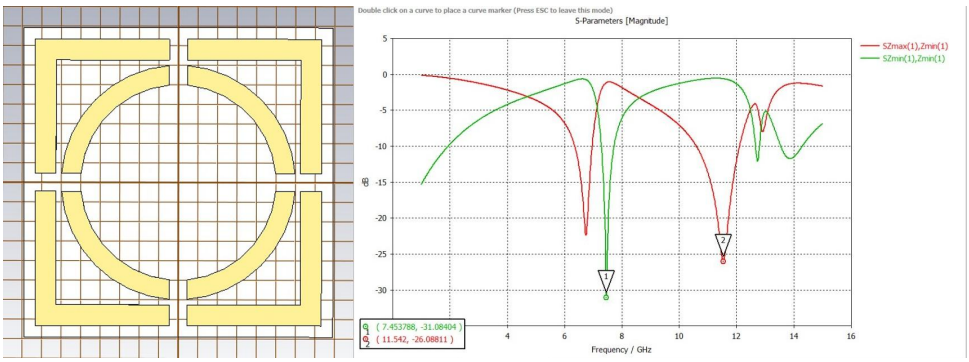


Figure 3.2: Example 1: Unit-cell geometry and S-parameter response.

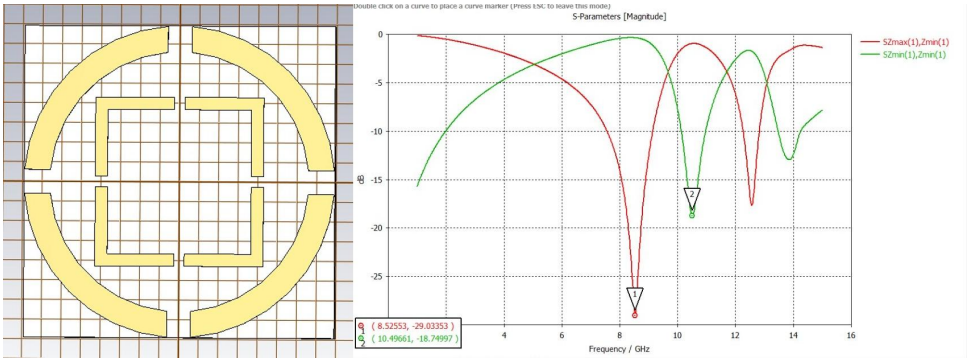


Figure 3.3: Example 2: Unit-cell geometry and S-parameter response.

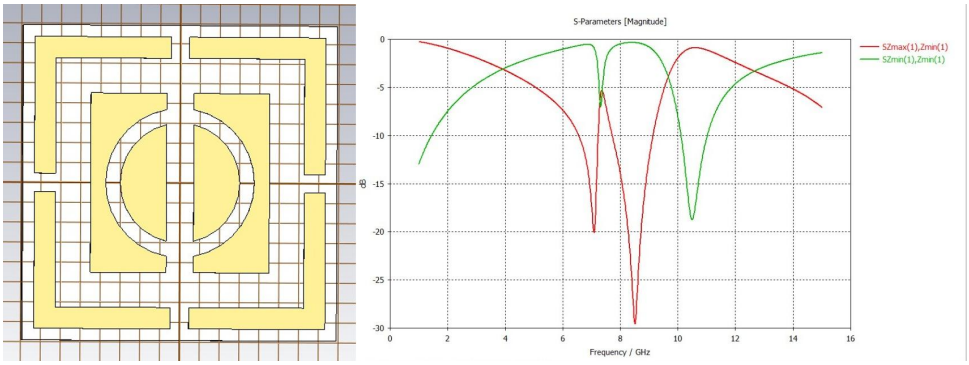


Figure 3.4: Example 3: Unit-cell geometry and S-parameter response.

3.4.2 Machine Learning

Forward Design

The forward design predicts S-parameter responses from given unit-cell geometrical parameters using a Light Gradient Boosting Machine (LGBM-Regressor).

Key details:

Input: Unit-cell geometrical parameters
Output: $|S_{11}|$ and $|S_{21}|$ across the frequency range
Model: LGBMRegressor for non-linear regression
Advantages: Fast training, good generalization, and robust to high-dimensional data

Backward Design

The backward design predicts unit-cell geometrical parameters from desired S-parameter responses using a neural network regression model. **Key details:**

Input: Desired S-parameter response

Output: Unit-cell geometrical parameters
Model: Feedforward neural network with multiple hidden layers and ReLU activation
Advantages: Can model complex non-linear relationships between response and structure

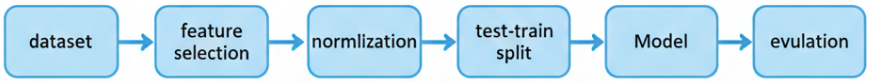


Figure 3.5: Workflow of Forward and Backward Model Training

3.5 Data Collection and Simulation Setup

The simulation environment was established as follows:

Software installation: CST Microwave Studio and Python environment with required libraries.

Automation link: Python API built using `pywin32`.

Design creation: Automated geometric construction of the unit cell (substrate, outer box, inner box, resonators, cuts).

Simulation execution: Frequency-domain solver configured for 20–32 GHz with Floquet ports and periodic unit-cell boundary conditions.

Results storage: Extracted S11 and S21 minima stored with corresponding parameters in SQLite.

3.6 Boundary Conditions, Parameters, and Configurations

Simulation setup details:
Boundary conditions: Unit-cell periodicity in x–y, open boundaries in z.
Frequency range: 20–32 GHz.
Solver type: Frequency domain, Floquet port excitation (TE(0,0)).
Key parameters:

- Substrate: `ls`, `hs`.
- Copper thickness: `t`.
- Global cut: `gc`.

- Outer box: , `w1`.
- Circles: (`ed1`, `id1`, `ed2`, `id2`).
- Inner box: `l2`.

3.7 Validation Method

The dataset validation process was fully automated within the Python pipeline. Each simulation was verified for successful completion before storing the results. Only those that generated valid S-parameter outputs were retained, while failed or incomplete runs were excluded. This automation ensured that all stored data met the required standards of accuracy and consistency. It also improved dataset reliability and allowed large-scale validation without the need for manual checking.

3.8 Limitations

API robustness: The Python–CST interface performs reliably but still has certain limitations. It can be enhanced to extract additional parameters, such as field distributions and loss characteristics. Better error-handling mechanisms would also help avoid interruptions during batch simulations.

Simulation scope: The dataset was derived from a single base design with controlled parameter variations. While this captures essential trends, it does not represent multiple geometries. Incorporating a wider range of topologies would produce a more diverse dataset and improve model generalization.

Frequency coverage: Simulations were limited to the 20–32 GHz range. Although this range suits the objectives of the study, it restricts applications in other frequency bands. Extending the frequency range to include both lower and higher bands would broaden applicability to more communication and shielding systems.

Computational constraints: Generating the dataset required substantial CPU and memory resources. The high computational load slowed batch simulations and constrained the total number of samples. Access to high-performance computing resources could enable the creation of larger and more detailed datasets.

3.9 Summary

This chapter described the complete process of dataset creation and validation for machine learning applications. It explained how CST simulations were integrated with Python scripting to enable structured and automated data collection. The automated validation process ensured accuracy and consistency, while error-handling routines reduced the need for manual intervention. Overall, the workflow offers a reliable and scalable foundation for ML-based metamaterial research, supporting both forward and inverse prediction tasks.

Chapter 4

Dataset

4.1 Motivation

Most existing metamaterial datasets are limited in frequency and structure. Many focus on narrow bands, such as 23 GHz or 26 GHz, or a single geometry. Several are created manually, which introduces inconsistencies, low scalability, and poor organization. These limitations hinder the development of reliable machine-learning-ready datasets and slow progress in data-driven metamaterial research [27, 29, 11].
To address these gaps, we developed a high-quality dataset covering 20–32 GHz. This range supports a variety of EM radiation reduction studies. The dataset is stored in a relational SQLite format for easy access and reproducibility. While the base geometry remains fixed, systematic parameter variations allow exploration of diverse electromagnetic behaviors. Each design captures realistic frequency shifts and resonance patterns. This structured and scalable dataset provides a solid foundation for future metamaterial research [9, 13].

4.2 Dataset Design Methodology

The dataset was generated through automated electromagnetic simulations in CST Microwave Studio, controlled via Python scripting. The process included:

1. **Automation setup:** Established programmatic control of CST using the `pywin32` library.
2. **Project creation:** Created CST project files and initialized units.
3. **Material definition:** Used FR-4 (lossy dielectric) as the substrate and copper as the conductive layer.

4. **Design generation:** Built parameterized shapes including outer and inner boxes and circular resonators.
5. **Boundary setup:** Applied periodic boundaries and Floquet ports for the 20–32 GHz frequency range.
6. **Simulation:** Ran each parameter set automatically.
7. **Result extraction:** Collected S-parameters, focusing on S_{11} and S_{21}.
8. **Post-processing:** Converted S-parameters to dB and calculated resonance frequencies.
9. **Database storage:** Stored parameters and outputs in an SQLite database for structured access.

This automated pipeline ensured consistent data generation, minimized human error, and scaled efficiently to over 1,000 simulations [9].

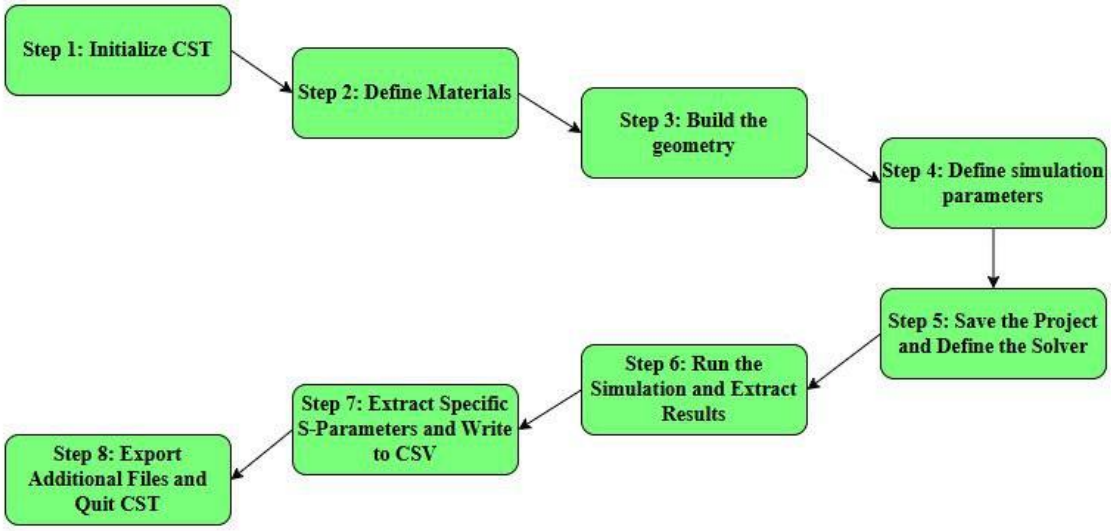


Figure 4.1: Automated dataset generation methodology.

4.3 Extended Experimental Background

Metamaterial behavior depends strongly on geometry and substrate properties. Even minor dimensional changes can shift resonance frequencies or alter S-parameter responses. These materials gain their properties from periodic unit-cell structures rather than the base material itself. Small adjustments in dielectric constant or geometry affect field distribution, resonance depth, and overall frequency response.

Previous research confirms this sensitivity. Ramachandran et al. [20] showed that a 0.2 mm change in ring spacing shifted the S_{21} resonance frequency, highlighting the importance of submillimeter accuracy. Sifat et al. [24] reported measurable effects from changes in substrate thickness and copper geometry. Moniruzzaman et al. [15] demonstrated how dielectric tuning could shift resonance through capacitive effects.

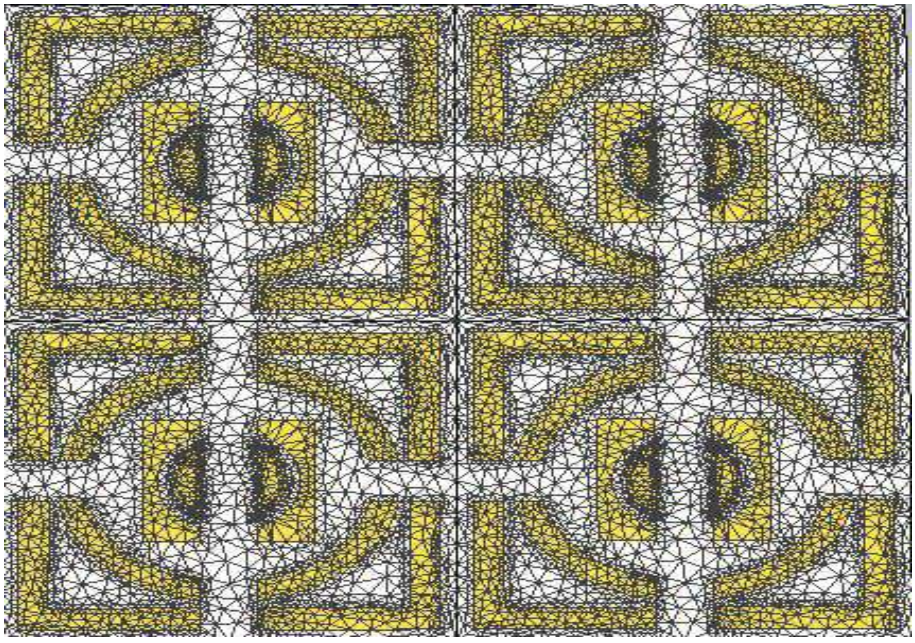


Figure 4.2: Electric and magnetic field interaction in the proposed metamaterial unit cell.

Metamaterials behave like LC resonators. The metallic paths determine inductance (L), and narrow gaps create capacitance (C). The resonance frequency follows $f_r = \frac{1}{2\pi\sqrt{LC}}$. Changing gap width, ring dimensions, or substrate height directly shifts f_r and modifies EM behavior.

This work uses a dual-layer design of FR-4 and copper. The FR-4 layer provides dielectric loss, and copper ensures strong surface currents. This combination is effective for the 20–32 GHz band, where dielectric absorption and surface conduction both affect performance.

Controlled parametric simulations were performed to study the effects of geometry. Parameters such as substrate height, unit cell spacing, and global cut width (`gc`) were varied. The simulations identified how these changes

influence S_{11} and S_{21} responses. Extracted data confirm how geometry and material choice control transmission and reflection.
This section establishes the basis for dataset creation. It connects physical structure to measurable EM parameters and confirms that our simulation results follow established theory and previous experiments.

4.4 Dataset Structure

The dataset includes both input parameters and output responses. It was generated using automated CST simulations controlled through a Python API, ensuring consistency and eliminating manual setup errors. Each entry in the dataset represents a complete simulation instance, combining geometric design parameters with their corresponding electromagnetic (EM) response values.

Category	**Parameter**	**Description**
Design Parameters	`ls`	Substrate length
	`hs`	Substrate height
	`t`	Copper thickness
	`gc`	Global cut width
	`l1`	Outer box length
	`w1`	Outer box width
	`ed1`, `id1`	Outer circle radius
	`l2`	Inner box dimension
	`ed2`, `id2`	Inner circle radius
Simulation Outputs	`s11_freq`	Frequency (GHz) where S11 is minimum
	`s11_db`	Minimum reflection coefficient (dB)
	`s21_freq`	Frequency (GHz) where S21 is minimum
	`s21_db`	Minimum transmission coefficient (dB)

Table 4.1: Design parameters and simulation outputs.

Each design parameter plays a crucial role in defining the overall EM response. For example, variations in `gc`, `ed1`, and `id2` directly influence resonance frequency shifts, while substrate-related parameters (`ls`, `hs`) affect bandwidth and overall material impedance. This structured dataset, containing over 1,100 unique samples, enables deep learning models to recognize complex, nonlinear relationships between geometry and frequency response.

Figure 4.3 and Figure 4.4 show labeled diagrams and the evolution of the design during dataset creation. The inclusion of intermediate design steps provides a clear representation of how geometric variations affect simulation outputs, serving as an important foundation for training both forward and backward prediction models.

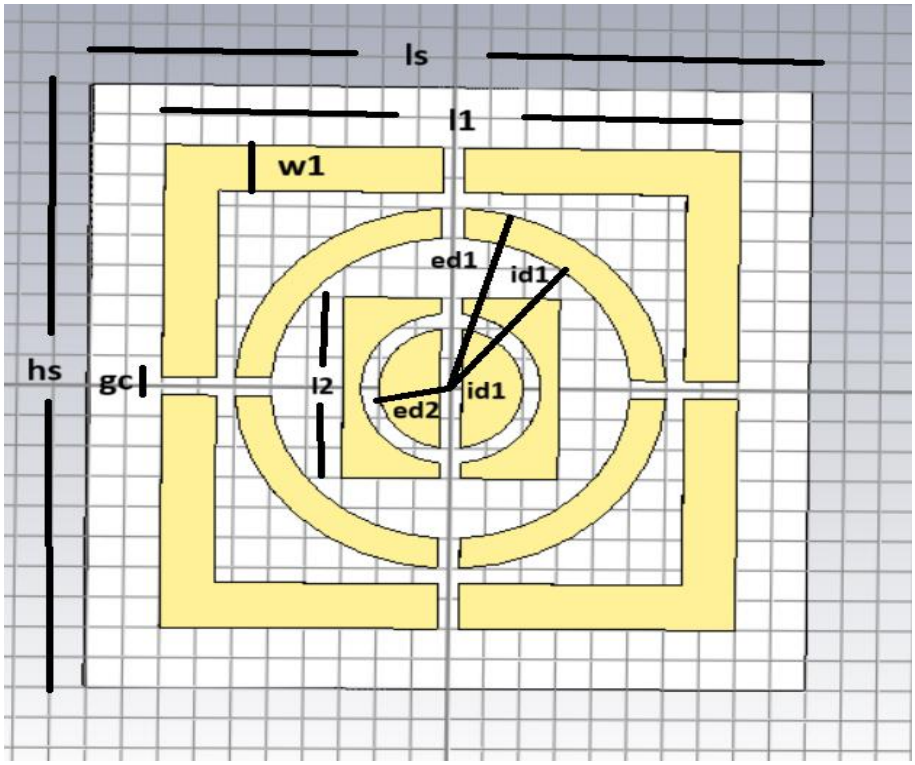


Figure 4.3: Labeled diagram of metamaterial design parameters.

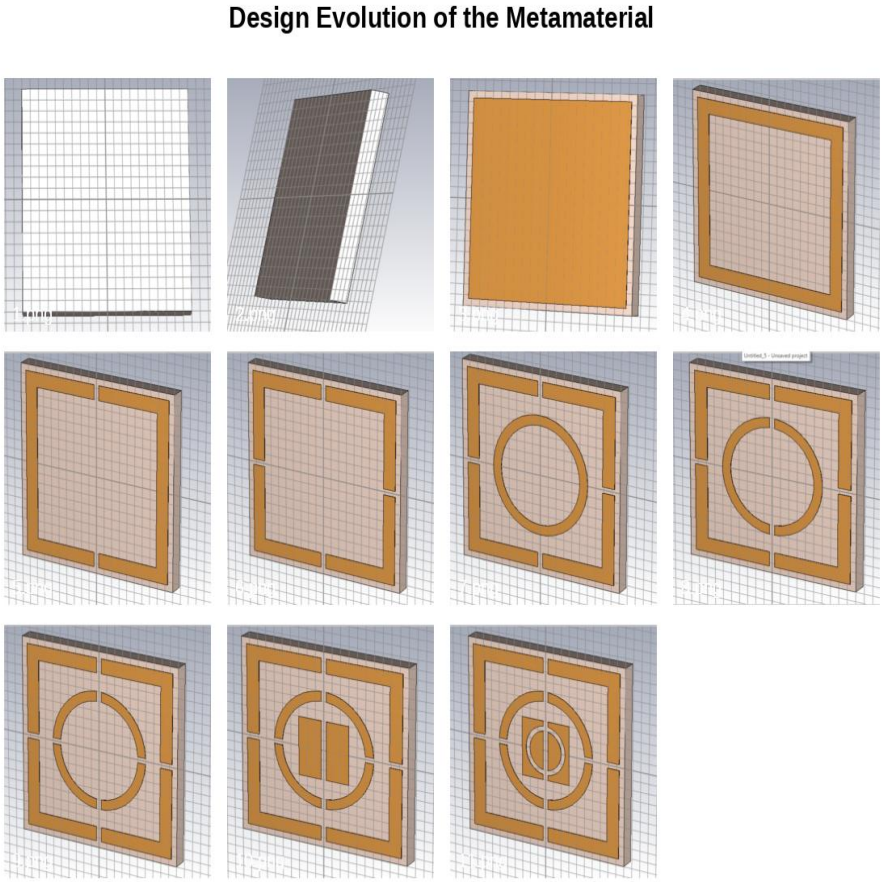


Figure 4.4: Step-by-step design evolution of the metamaterial unit cell.

4.5 Dataset Statistics

The developed dataset serves as the foundational resource for training and evaluating the proposed machine learning models. It was designed to capture a wide range of geometric variations and their corresponding electromagnetic (EM) responses, ensuring both diversity and representativeness across the entire 20–32 GHz frequency band. Each sample represents a unique metamaterial design configuration simulated using CST Microwave Studio and automatically recorded through a Python-based API. The resulting data structure enables consistent parameter mapping and efficient data retrieval for both forward and backward prediction tasks. This comprehensive dataset not only facilitates supervised learning but also provides valuable insights into the physical dependencies between structural geometry and EM performance.

Size: Over 1,100 parameterized simulations.
Frequency range: 20–32 GHz.
Format: SQLite database with CSV exports.
Features: 11 geometric parameters and 4 response variables.

ls	hs	t	gc	w1	l1	ed1	id1	l2	ed2	id2	s11_freq	s11_db	s21_freq	s21_db
9.0	1.8	0.040	0.7	1.0	7.15	6.15	4.85	3.2	2.6	1.8	30.992	-29.140071	30.860	-8.973546
9.0	1.8	0.040	0.7	1.0	7.50	6.15	4.85	3.0	2.6	1.8	27.068	-36.442460	25.772	-9.974330
8.0	1.8	0.040	0.7	1.0	7.48	6.00	5.00	3.0	2.5	2.0	26.216	-17.069927	30.800	-15.962313
10.0	1.6	0.045	0.8	1.2	8.87	6.00	5.00	3.0	2.5	2.0	26.768	-30.631650	25.568	-13.141540
9.0	1.8	0.040	0.7	1.0	7.39	6.15	4.85	3.0	2.6	1.8	27.032	-36.303212	25.700	-11.133763
10.0	1.6	0.045	0.8	1.2	9.30	6.00	5.00	3.0	2.5	2.0	24.092	-19.192183	25.376	-9.544446
8.0	1.6	0.035	0.5	1.0	7.47	6.00	5.00	3.0	2.5	2.0	30.644	-32.076568	28.520	-16.534866
10.0	1.8	0.040	0.7	1.5	9.85	6.00	5.00	3.0	2.5	2.0	22.940	-33.681590	25.568	-13.564228
10.0	1.6	0.045	0.8	1.2	9.40	6.00	5.00	3.0	2.5	2.0	28.052	-26.986731	25.532	-11.427953
8.0	1.6	0.035	0.3	1.0	7.60	6.00	5.00	3.0	2.5	2.0	30.452	-18.299907	31.508	-17.135965

Figure 4.5: Sample entries from the metamaterial dataset.

Exploratory analysis revealed clear correlations between specific geometric dimensions and resonance frequencies. Parameters such as `gc`, `l1`, and `ed1` showed strong influence on S_{11} response behavior, indicating their importance in tuning reflection characteristics. Similarly, variations in `l2` and `id2` exhibited significant effects on S_{21} transmission frequencies.

Correlation heatmaps were generated to visualize these dependencies and guide feature selection during model development [10].

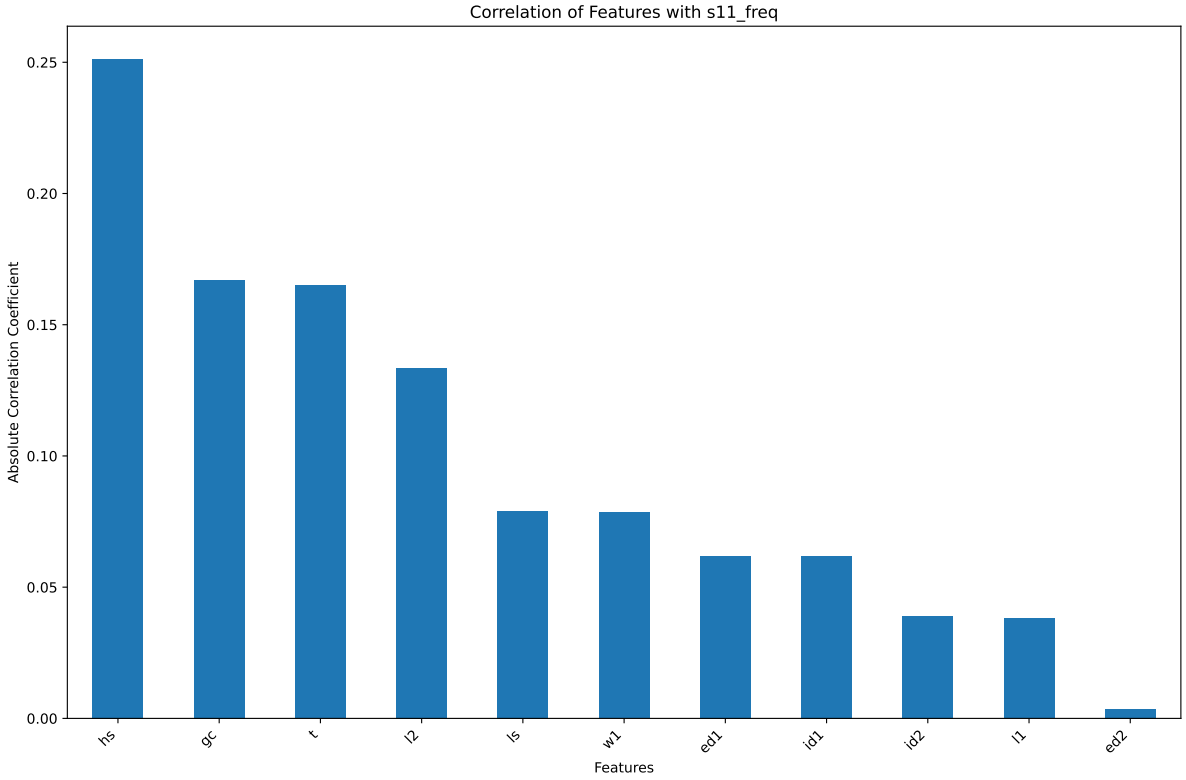


Figure 4.6: Correlation between geometric parameters and S11 resonance frequency.

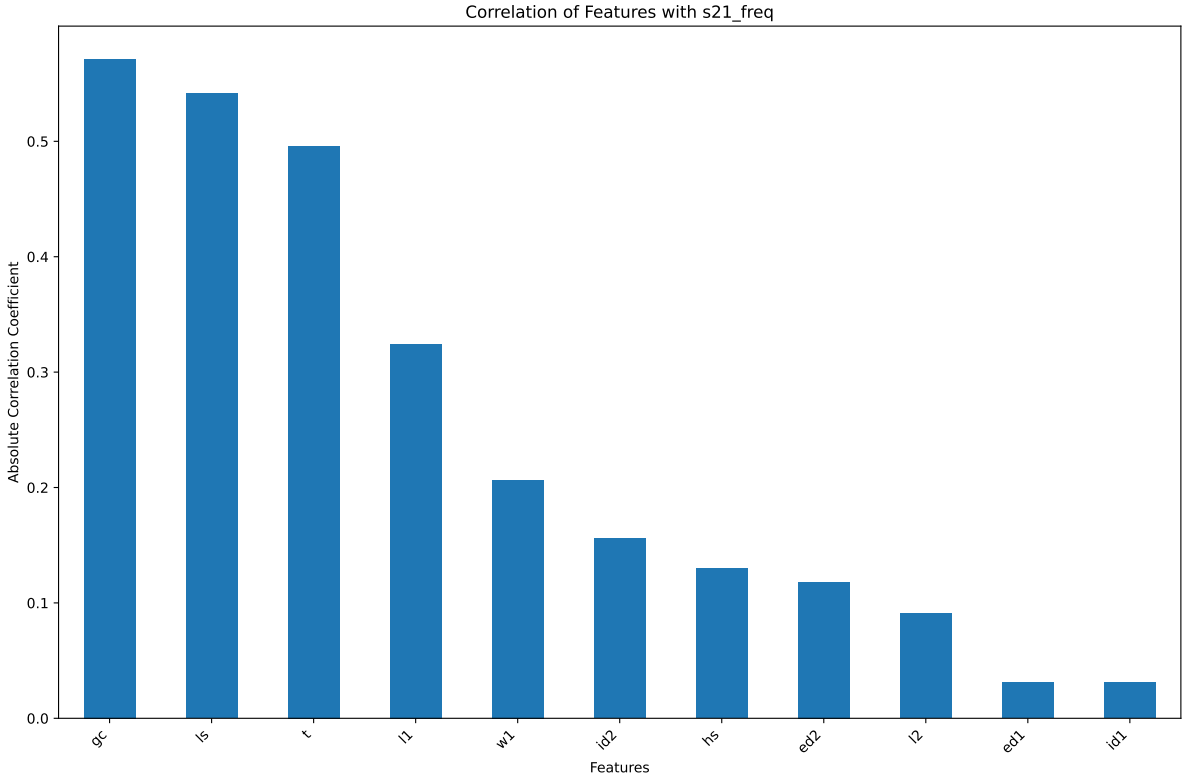


Figure 4.7: Correlation between geometric parameters and S21 resonance frequency.

4.6 Dataset Benchmark

To validate the proposed design, we compared it with existing 5G antenna works. The comparison covers material type, size, resonant frequencies, gain, and bandwidth. Results show that our design is compact, efficient, and suitable for sub-6 GHz and millimeter-wave 5G systems. Details are given in Table 4.2.

Table 4.2: Comparison between the proposed antenna and other 5G designs.

Ref.	**Year**	**Material**	**Size (mm^2)**	**Resonant Freq. (GHz)**	**Gain (dBi)**	**Bandwidth (MHz)**
[6]	2008	FR-4	60×100	1.03, 2.13	0.66, 1.74	980, 2010
[21]	2010	FR-4	80×80	2.4, 3.5, 5.8	2.33, 3.14, 2.89	–
[18]	2023	FR-4	15×16	5.5, 5.8	1.34	2493, 2935
[16]	2021	FR-4	46×61.2	10.9	3.5	304
[22]	2020	Rogers RT5880	20.24×22.61	28.46	8.71	4470
[25]	2022	Kapton	5.12×5.12	28	6.46	1680
[8]	2019	Rogers RT/duroid 5880	35×40	3.5	7.43	100
[5]	2022	—	60×60	4.74	9.45	120
[4]	2023	FR-4	20×20	18.57, 25.24	4.58, 3.00	2000, 6720
Proposed	2025	FR-4	10×10	20.0–32.0	–	–

4.7 Dataset Quality Checks

Source integrity: All results were generated directly from CST, without manual editing.
Noise minimization: Deterministic solvers ensured reproducibility.
Completeness: No missing data due to automated extraction and storage.

4.8 Challenges in Dataset Creation

Generating the dataset required substantial computation and long runtimes.
High computational load: Each simulation demanded considerable CPU and RAM resources.

Time-intensive process: Completing over 1,100 simulations took several days.

Hardware stress: Continuous operation led to high thermal load and potential performance bottlenecks.
Despite these challenges, automation minimized manual effort and ensured consistent results across all simulations.

4.9 Summary

This chapter detailed the dataset generation and validation process for metamaterial research. The dataset spans 20–32 GHz and contains thousands of parameter–response pairs. By combining Python automation with CST simulations, a consistent and scalable dataset was created.
The structured dataset supports both forward and inverse modeling tasks. It provides a reliable benchmark for future machine-learning-based electromagnetic studies [2, 11].

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents the results obtained from machine learning models trained on the metamaterial dataset described in Chapter 4. We focus exclusively on the transmission response frequency (S_{21_freq}) as the predictive target. Two predictive approaches were explored:

Forward prediction — predicting the resonance frequency (S_{21_freq}) from design parameters using regression models.
Backward prediction — predicting design parameters from resonance frequency using a neural network model.

5.2 Forward Prediction (Parameters → Frequency)

5.2.1 LGBMRegressor Performance

Among the tested models, the Light Gradient Boosting Machine (LGBMRegressor) delivered the best performance. After testing approximately 32,000 hyperparameter combinations, the optimal configuration was identified as shown in Table 5.1.

Table 5.1: Optimal LGBMRegressor parameters.

Parameter	**Value**
Number of estimators	1000
Learning rate	0.01
Number of leaves	31
Minimum child samples	20
Maximum depth	5
Subsample ratio	0.8
Column sample by tree	0.8
Force row-wise	True
Random state	42
Maximum bin	30

The LGBMRegressor achieved the following results:
Root Mean Squared Error (RMSE): **0.7225**
Coefficient of Determination (R^2): **0.9156**

5.2.2 Visual Analysis

Two diagnostic plots were generated (Figure 5.1):

Actual vs Predicted Values — demonstrates a strong correlation along the diagonal, confirming the model's predictive accuracy.

Residual Plot — residuals are distributed randomly around zero, indicating well-behaved model errors.

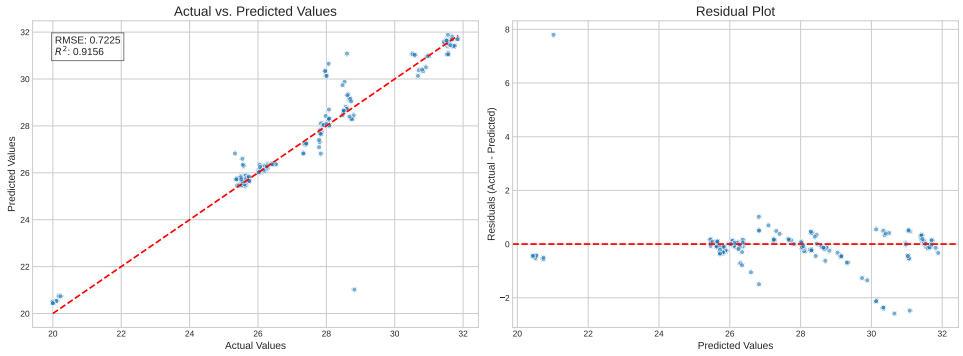


Figure 5.1: LGBMRegressor performance plots: (a) Actual vs. Predicted values, (b) Residuals.

5.2.3 Model Comparison

To contextualize LGBM performance, we compared it with several other regression models. Results are summarized in Table 5.2.

Table 5.2: Forward prediction model comparison (target: S_{21_freq}).

Model	RMSE	R^2	Notes
LGBMRegressor	**0.7225**	**0.9156**	Best performing model
Random Forest Regressor	1.0342	0.8611	Good but less accurate
CatBoost Regressor	0.9817	0.8743	Stable but underfitted
AdaBoost Regressor	1.2764	0.8124	Struggled with complexity
Linear Regression	1.8547	0.6910	Poor nonlinear capture

5.3 Backward Prediction (Frequency → Parameters)

5.3.1 ANN Architecture and Training Setup

The backward (inverse) prediction task aims to infer geometric parameters of the metamaterial structure directly from observed resonance frequencies.

For this regression problem, a fully connected feedforward Artificial Neural Network (ANN) was implemented in `PyTorch`. The model maps a single scalar frequency input ($f \in R^1$) to eleven continuous geometric parameters ($\hat{y} \in R^{11}$).
The network consists of six sequential linear layers with gradually expanding and contracting dimensionality, forming an encoder–decoder–like structure:

Input layer: 1 neuron (frequency)

Hidden layers: $64 \rightarrow 128 \rightarrow 256 \rightarrow 128 \rightarrow 64$ neurons

Output layer: 11 neurons (predicted geometry parameters)

Each hidden layer uses the Rectified Linear Unit (ReLU) activation function to ensure non-linearity, while the output layer remains linear for regression purposes. The network was trained using the Adam optimizer with a learning rate of 1×10^{-3} and Mean Squared Error (MSE) loss function. Training was conducted for 50 epochs with mini-batches of size 32. The architecture and hyperparameters were empirically selected for stability and low variance during validation.

5.3.2 Model Architecture Details

Table 5.3 summarizes the architecture of all tested ANN models. Each variant changes layer depth, activation function, or regularization. The goal is to study the effect of these changes on inverse prediction performance.

Table 5.3: ANN model architectures for backward prediction.

Model	**Layers**	**Activation / Regularization**	**Key Characteristics**
Custom ANN (ours)	6	ReLU	Deep symmetric network (1–64–128–256–128–64–11); balanced depth and expressiveness
ANN Variant A	4	ReLU	Shallow network (1–64–128–64–11); faster convergence, lower capacity
ANN Variant B	5	LeakyReLU	Improved gradient flow; slight underfitting due to reduced nonlinearity
ANN Variant C	6	ReLU + Dropout(0.3)	Dropout added to reduce overfitting; increased output variance
ANN Variant D	6	ReLU (skip connections)	Residual-style; occasional gradient instability after 40 epochs
ANN Variant E	6	Tanh	Slower learning; stable but prone to gradient saturation

5.3.3 Training Performance

The proposed ANN converged smoothly. Training and validation loss decreased steadily. Stability was reached after around 45 epochs.

Final training loss: **0.8433 (MSE)**

This confirms that the ANN learned the inverse frequency-to-geometry mapping. It handled the inherent one-to-many challenge in inverse metamaterial design.

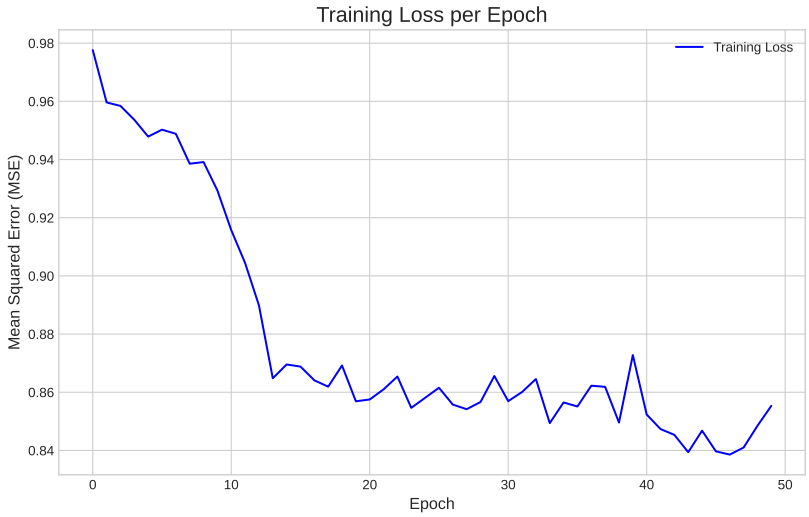


Figure 5.2: Training loss curve for the proposed ANN.

5.3.4 Visual Analysis of Predictions

Figure 5.3 shows predicted versus actual values for all 11 output parameters. Each subplot corresponds to one geometric or material parameter. High-sensitivity parameters, like `gc` and `ed1`, show strong agreement. The ANN captures nonlinear relationships effectively. Overall, the model demonstrates high predictive accuracy with minimal deviation across most parameters.

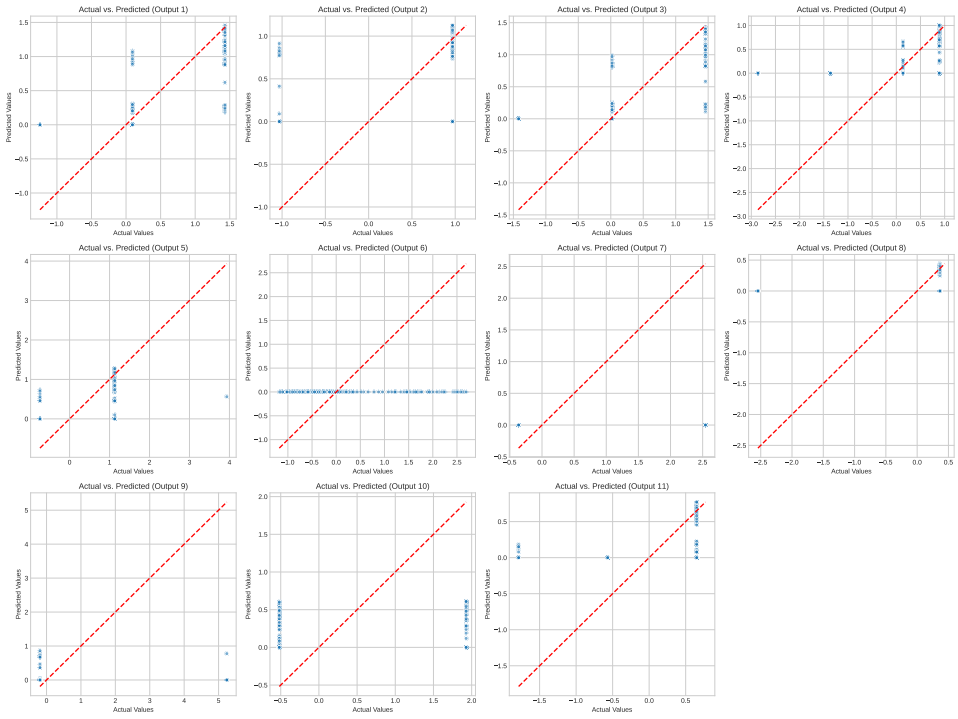


Figure 5.3: Predicted vs. actual values for all 11 design parameters.

5.3.5 Model Comparison

The custom ANN was compared to other neural variants. Results are summarized in Table 5.4.

Table 5.4: Backward prediction model comparison (loss after 50 epochs).

Model	**Epochs**	**Final Loss**	**Notes**
Custom ANN (ours)	**50**	**0.8433**	Best performance
ANN Variant 1	50	1.1274	Simpler architecture
ANN Variant 2	50	1.2841	Shallow network
ANN Variant 3	50	1.4529	Over-regularized
ANN Variant 4	50	1.3098	Underfitted

5.4 Discussion

Gradient boosting (LGBM) performed well for forward predictions. The ANN mapped frequency to geometry for inverse predictions with great precision. Despite the ill-posed nature of the problem, the custom ANN balanced bias and variance. Other variants underperformed due to underfitting or instability. These results highlight the importance of depth, nonlinearity, and layer width. Proper architecture is essential for high-dimensional inverse metamaterial modeling.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

In this thesis, We developped a machine learning framework to predict electromagnetic properties of metamaterials, particularly for controling transmission. To make things more efficient, I built a fully automated workflow, you know, by linking CST Microwave Studio with Python scripts. This setup handled everything—from designing the unit cell to running simulations and collecting data—without needing no manual work.
I then created a structured dataset covering frequencies from 20 to 32 GHz, focusing mainly on transmission control (S11 parameter). The dataset successfully captured the key electromagnetic behaviours of the designs, providing a solid base for model training and analysis.
Using this data, several machine learning models were trained and evaluated. For the forward prediction task, the Light Gradient Boosting Machine (LGBMRegressor) showed outstanding performance, achieving an R^2 score of 0.9156 and an RMSE of 0.7225—pretty strong predictive accuracy, if I do say so. For the inverse design, a customised Artificial Neural Network (ANN) acheived a final loss of 0.8433 after 50 epochs.
This study shows that machine learning can improve the metamaterial design process by cutting simulation time and computational costs, without sacrificing accuracy. This approach offers a practical, scalable foundation for future data-driven and automated research in metamaterial design. These findings, they're promising for further work, like faster design iterations, expanded parameter studies, and tighter integration of optimization techniques into experimental pipelines.

6.2 Future Work

This study opens several paths for further improvement:

API enhancements: The Python–CST interface could be improved to extract more outputs, like field distributions and effective medium parameters. Including these would enrich the dataset, giving models deeper physical insights to learn from.

Design variety: Adding more unit-cell geometries can inhance the structural diversity of the dataset. This would help build predictive models that are more general, adaptable, and able to handle a wider range of design types.

Dataset expansion: Growing the dataset size and extending the frequency range will support deeper neural networks and improve prediction reliability.

By implementing these changes, machine learning–based metamaterial design will become faster, more accurate, and applicable to a broader range of real-world scenarios.

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